

Solar energy battery design assignment

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Abstract

Photovoltaic (PV) systems are a renewable energy technology widely used for residential and commercial power generation, grid-connected systems, and large-scale infrastructure such as data centres and artificial intelligence (AI) systems. PV technology represents a pathway to energy independence by reducing reliance on conventional, mass-market fossil-fuel-based power generation. In systems with multiple PV panels connected in series, the output voltage is often higher than required, making DC-DC buck converters essential for stepping down voltage and enabling efficient energy transfer. The cost of PV arrays has decreased in recent years due to advances in manufacturing and investment. Further progress in research and development enables continuous advancement in PV technology.

Introduction

In real-world applications, PV modules are connected in both series and parallel to achieve the required voltage and current levels. In arrays where multiple solar panels are connected, the output voltage is expected to exceed the battery voltage. Therefore, a DC-DC buck converter in Fig.1 is implemented to step the voltage down to an optimal power level for efficient battery charging. The converter operates by adjusting the duty cycle of an IGBT (insulated-gate bipolar transistor), thereby enabling efficient energy conversion. A typical PV system with a DC-DC converter utilises maximum power point tracking (MPPT) to extract as much power as possible from the process. PV arrays operate at the maximum power point (MPP), which is the optimal voltage for maximum power extraction. This is achieved via the perturb-and-observe (P&O) method, which perturbs the voltage toward the optimal point. This method is widely utilised due to its ease of use and low cost. Information taken from - [2]. The MPPT algorithm continuously adjusts the operating point of the PV array to maintain operation near the MPP under varying environmental and internal device conditions.

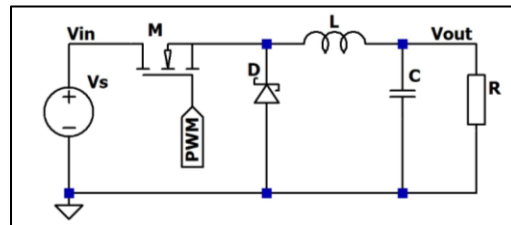


Figure.1 DC-DC buck converter. Picture taken from - [1]

One of the varying PV conditions is temperature, which influences the array's semiconductor properties. Since the array is mostly silicon (Si) which is a semiconductor material its bandgap energy decreases with an increase in temperature, meaning less energy required for an electron to be excited from the valence band to the conduction band (electron-hole pair), thus requiring less potential difference to maintain the separation of charge carriers across the p-n junction, resulting in a reduction in the open-circuit voltage (V_{oc}) and consequently in a decrease in the overall output of the array. The dependence of open-circuit voltage on temperature can be described by:

$$(1) V_{oc} = \frac{kT}{q} \ln \left(\frac{I_{sc}}{I_0} \right)$$

Formula taken from- [3], where k is the Boltzmann constant, T is the absolute temperature in Kelvin, q is the electron charge, and I_{sc} is the short-circuit current and I_0 is the saturation current, the formula illustrates that the open circuit voltage V_{oc} increases logarithmically with illumination current and decreases with increase in temperature, highlighting inverse relationship between temperature and photovoltaic output voltage.

This work presents the design and simulation of a PV battery charging system using MPPT via a DC-DC buck converter. The system in this simulation is designed to step down the PV array output voltage from 150 V to 48 V for efficient battery charging. This work covers PV array configuration and characterisation under varying irradiance and temperature conditions, a proportional-integral (PI) voltage controller, and P&O MPPT algorithm design, and IGBT power losses using the IKW30N60H3 datasheet. Simulation results are validated against theoretical calculations and MATLAB datasheet values for system accuracy.

Simulation Results and Discussion

I PV array system and configuration design

1.1) Question: Please design a suitable PV array system and configuration (number of modules and their connection) if the Green Technology A10J-M60-220 solar modules are employed. The PV array output voltage and output current are around 150 V and 15 A, respectively at the standard condition(solar irradiance of $1000 \text{ W} \times \text{m}^{-2}$, temperature of 25°C).

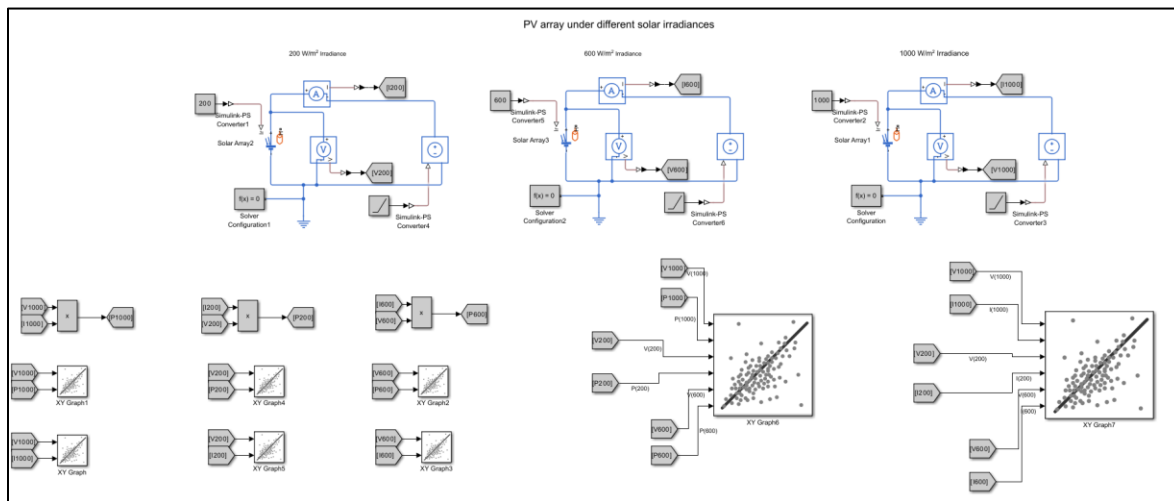


Figure 2 Simulink model of PV array for I-V and P-V characteristics under varying solar irradiance (200, 600, 1000 $\text{W} \times \text{m}^{-2}$).

1.2) Plot and present the I-V and P-V curves of the PV array under different solar irradiances: $S = 200 \text{ W} \times \text{m}^{-2}$, $600 \text{ W} \times \text{m}^{-2}$, and $1000 \text{ W} \times \text{m}^{-2}$, respectively. (The cell temperature remains at constant temperature of 25°C)

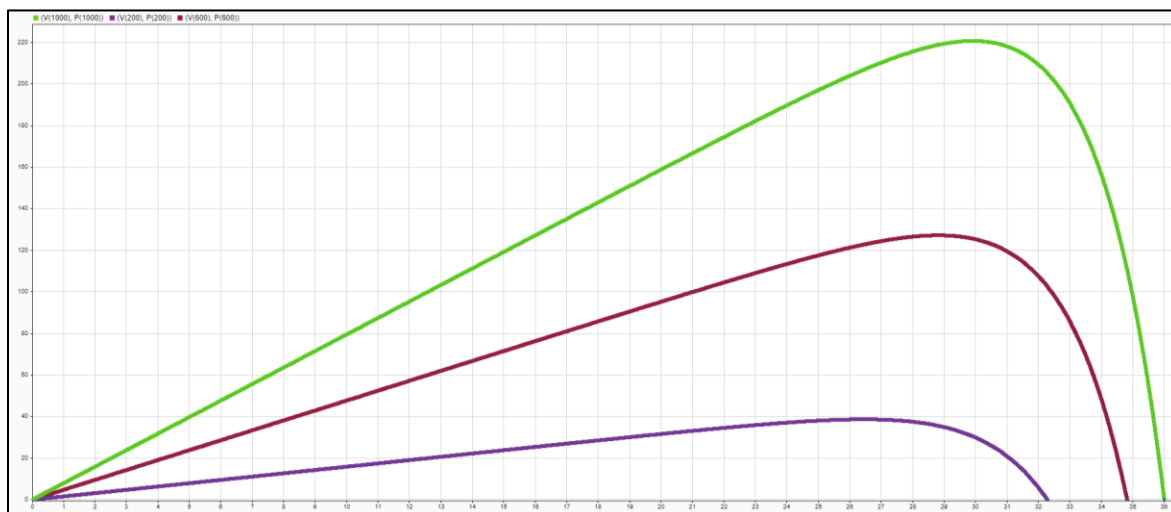


Figure 3 P-V characteristics of PV array at 200, 600 and 1000 $\text{W}\times\text{m}^{-2}$ (X-axis- Voltage, Y-axis- Power)

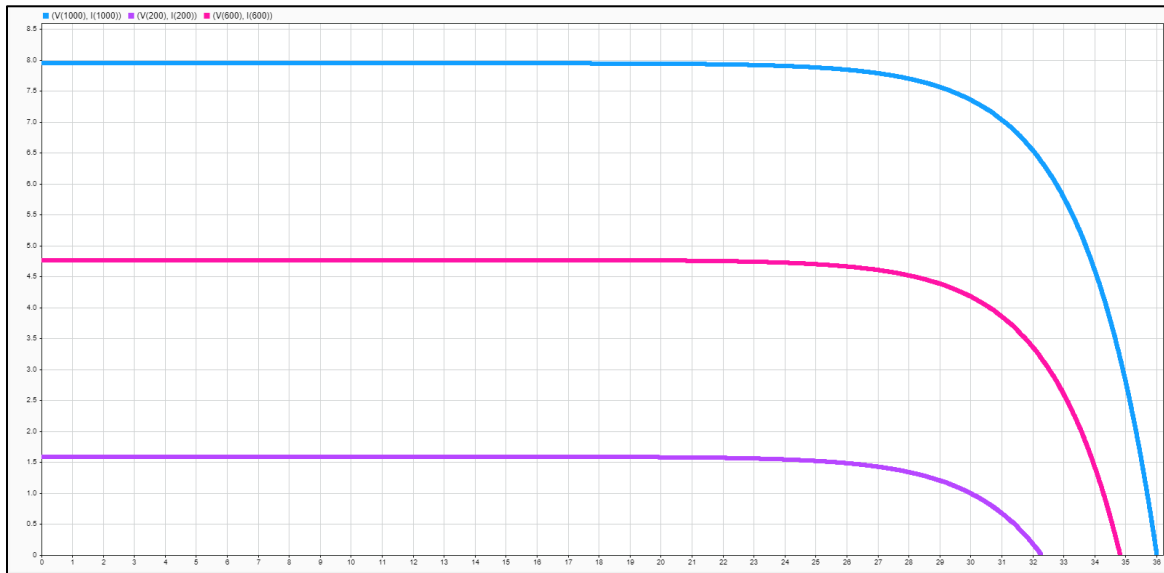


Figure 4 V-I characteristics of PV array at 200,600 and 1000 $\text{W}\times\text{m}^{-2}$ (X-axis- Voltage, Y-axis- Current)

2) Use the Data Cursor to record ***V_{oc}***, ***I_{sc}***, ***V_{max}***, ***P_{max}***, for each solar irradiance and fill Table-1.1.

Irradiation ($\text{W}\times\text{m}^{-2}$)	<i>I_{sc}</i> (A)	<i>V_{oc}</i> (V)	<i>V_{max}</i> (V)	<i>I_{max}</i> (A)	<i>P_{max}</i> (W)
200	1.59	32.28	26.53	1.46	38.67
600	4.77	34.82	28.82	4.41	127.20
1000	7.95	36.0	29.25	7.55	220.71

Table 1.1 recordings of ***I_{sc}***, ***V_{oc}***, ***V_{max}***, ***I_{max}***, and ***P_{max}*** under varying irradiation

3) Comment on the effect of solar irradiance on short circuit current ***I_{sc}*** and open-circuit voltage ***V_{oc}*** as well as ***P_{max}***. Present the ***P_{max}***-irradiance curve.

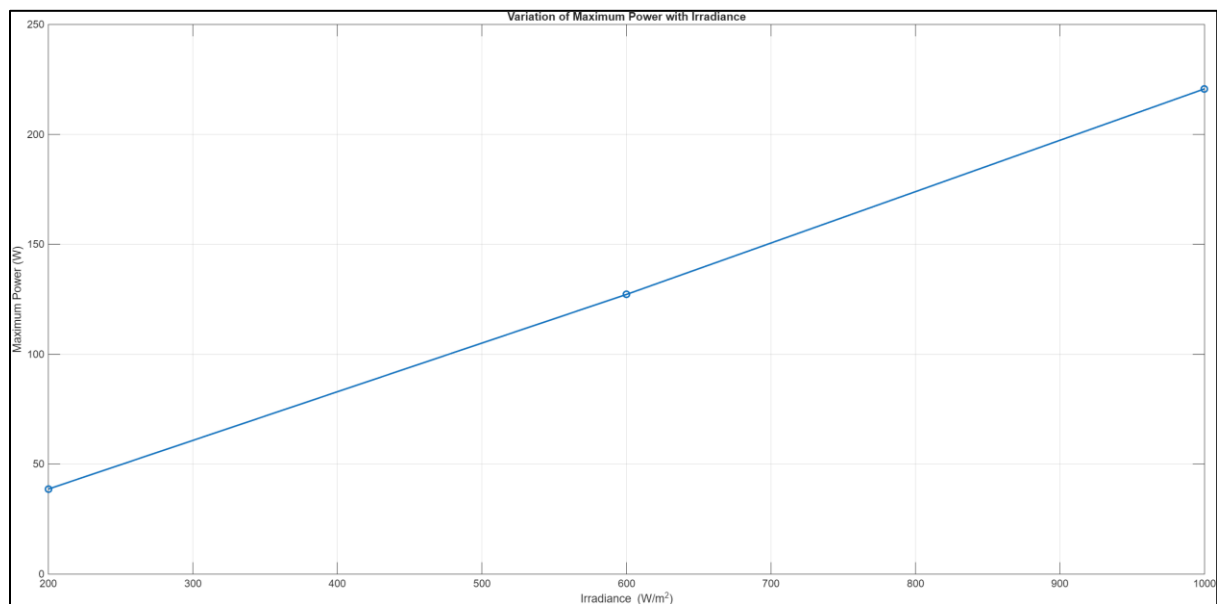


Figure 5: ***P_{max}*** - irradiance curve

Increase in solar irradiance forces I_{sc} to increase linearly, since the photocurrent that is generated by the PV cell is directly proportional to the incident light. In contrast, the open-circuit voltage V_{oc} increases only slightly with irradiance, since its dependence on irradiance is logarithmic. As a result, the maximum power P_{max} increases significantly with irradiance, since the current increases linearly while the voltage varies only slightly, resulting in an overall rise in power output.

1.3) Plot and present the I-V and P-V curves of the selected PV panel under different cell temperature: $T = 25^{\circ}\text{C}$, 45°C and 65°C , respectively. (The solar irradiance remains at constant irradiance of $1000 \text{ W}\times\text{m}^{-2}$).

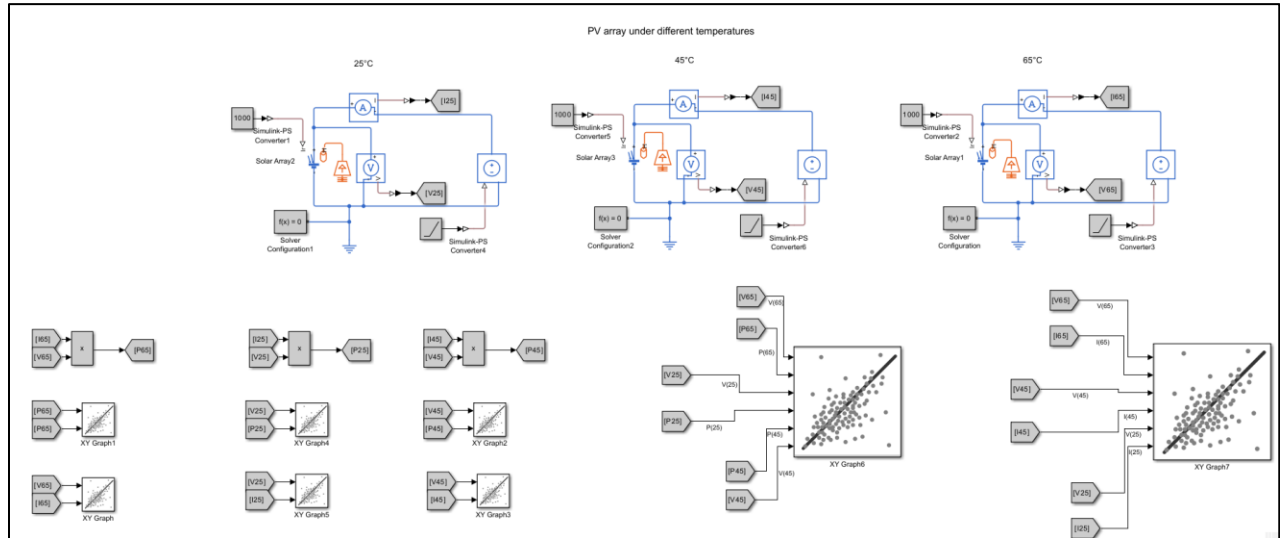


Figure 6 Modified MATLAB circuit under different temperatures (25°C , 65°C , and 45°C) but at a constant radiation of $1000 \text{ W}\times\text{m}^{-2}$

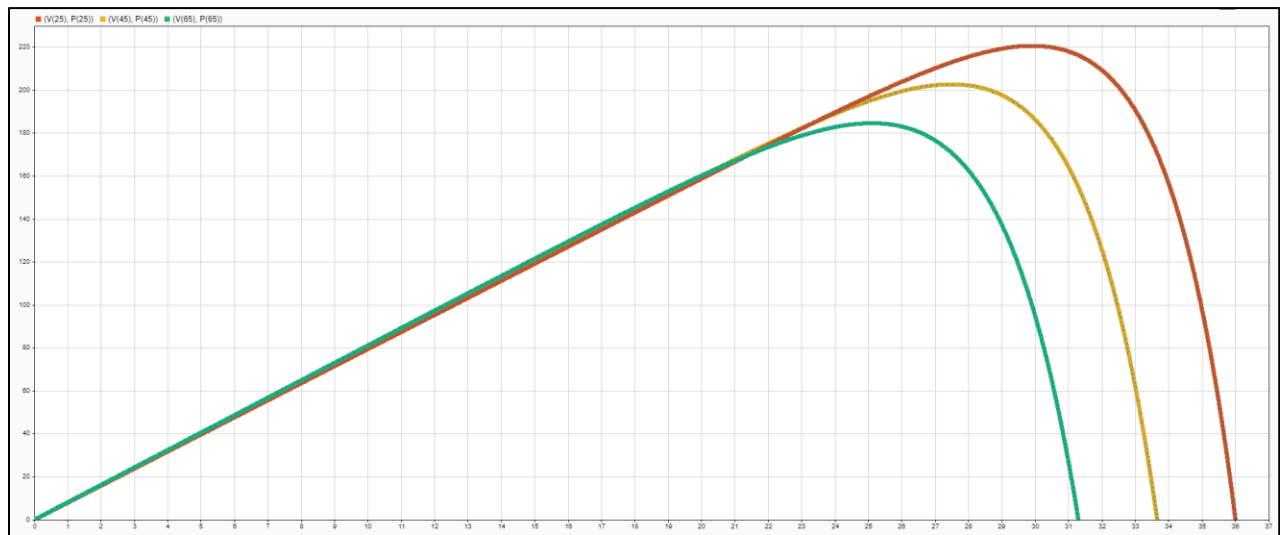


Figure 7 P-V characteristics of PV array at 25°C , 45°C and 65°C (x-axis: Voltage (V), y-axis: Power (W)).

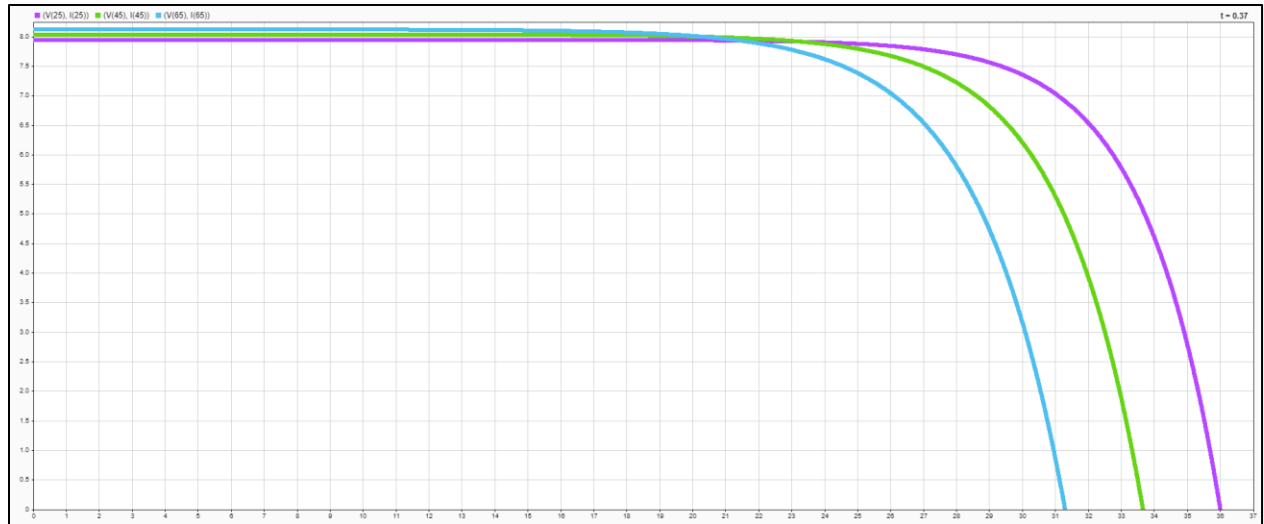


Figure 8 I-V characteristics of PV array at 25°C, 45°C and 65°C (x-axis: Voltage (V), y-axis: Current (A)).

2) Use the Data Cursor to record I_{sc} , V_{oc} , V_{max} , I_{max} , and P_{max} for each s and fill Table-1.2.

Temperature °C	I_{sc} (A)	V_{oc} (V)	V_{max} (V)	I_{max} (A)	P_{max} (W)
25	7.95	36.0	29.96	7.37	220.70
45	8.03	33.65	27.48	7.38	202.77
65	8.12	31.29	25.16	7.34	184.67

Table 1.2 recordings of I_{sc} , V_{oc} , V_{max} , I_{max} , and P_{max} under varying temperatures

3) Comment on the effect of temperature on short circuit current I_{sc} and open-circuit voltage V_{oc} as well as P_{max} , at constant irradiation of $1000(W \times m^{-2})$. Present the P_{max} temperature curve.

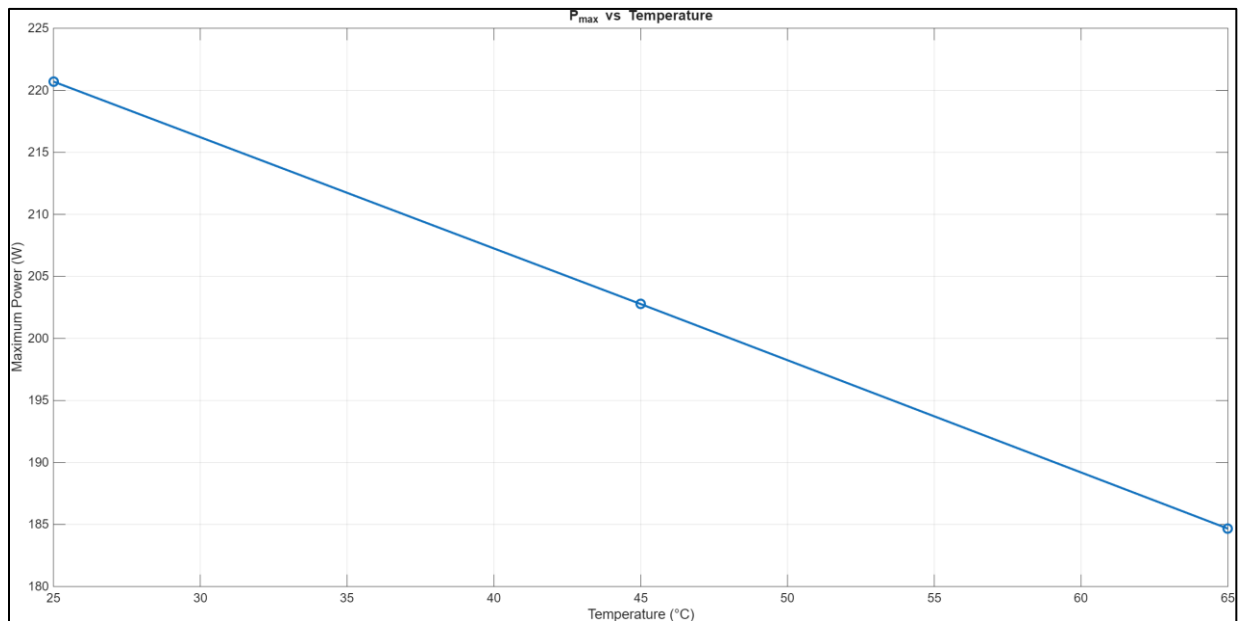


Figure 9. P_{max} temperature curve under varying temperatures

At a constant irradiance of $1000 W \times m^{-2}$, increasing temperature causes I_{sc} to increase slightly, from 7.95 A at 25 °C to 8.12 A at 65 °C, while open circuit voltage V_{oc} decreases with increase in temperature, from 36.0 V to 31.29 V. As a result, P_{max} decreases by 35 W from 220 W at 25 °C to 185 W at 65 °C. This indicates that higher temperatures may start with a slightly higher current

but ultimately lead to lower voltage and therefore lower power output. Hence, increasing temperature is suboptimal for MPP performance, since the available maximum power reduces, leading to a lower operating voltage point.

II Design and simulation of the solar energy battery charger circuit

2.1 1) Present the buck dc-dc converter circuit for the PV charger system with detailed control circuit as well as circuit description.

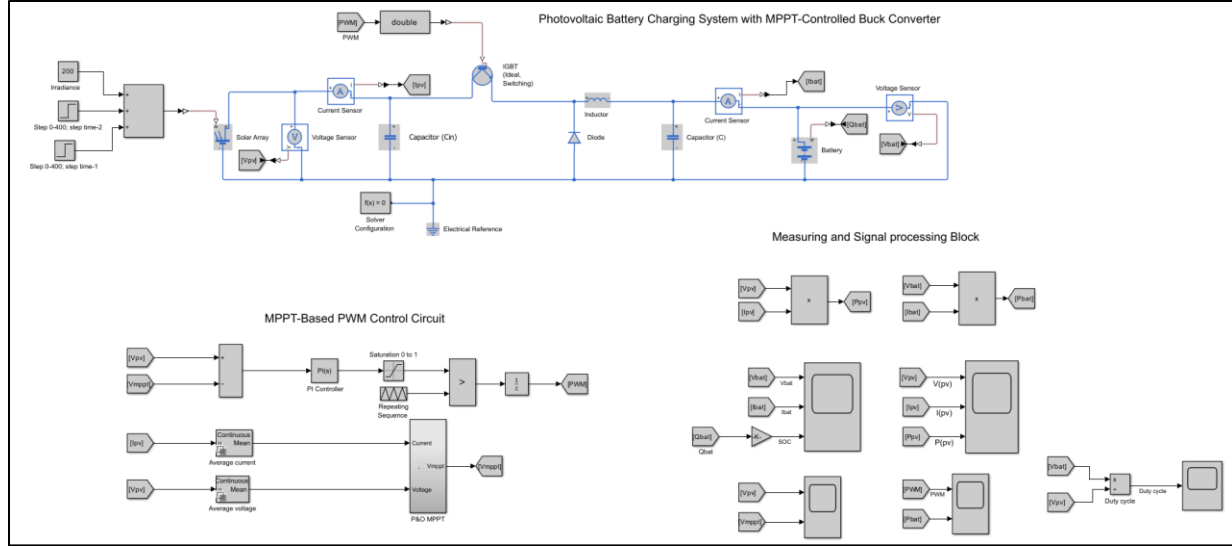


Figure 10. Buck DC-DC converter circuit for the PV charger system. Battery and solar array specifications are in Appendix A

Circuit components: PV array, IGBT (ideal switching), freewheeling diode (FWD), inductor (L), output capacitor (C), input capacitor (C_{in}), battery, ground.

Fig. 10 illustrates a photovoltaic battery charging system using an MPPT-controlled buck converter, including the power stage, control circuit, and measurement blocks. The PV array operates at 150.6 V and 14.6 A under standard conditions (1000 W/m^2 , 25°C), supplying a 48 V, 100 Ah battery with an initial state of charge of 50%. The buck converter operates at 20 kHz, with a maximum current ripple of 2 A over a 3-second charging period. The battery has an energy capacity of approximately 4.8 kWh (48 V, 100 Ah), making it suitable for small-scale energy storage applications such as solar systems or low-voltage electric mobility chargers. The 300 series cells per string were chosen to reach a threshold voltage of 150 V at MPP, since each cell produces around 0.5 V, meaning $300 \times 0.5 = 150 \text{ V}$ per string, with two parallel strings providing the required current of around 15 A.

The photovoltaic (PV) source is modelled as a solar array comprising two parallel strings, each with 300 solar cells in series. The PV array is designed to operate at a higher voltage than the battery, enabling efficient step-down conversion with a buck DC-DC converter. MPPT continuously adjusts the buck converter's duty cycle to ensure the PV array delivers maximum available power. Buck topology is employed because the battery requires a lower, regulated voltage for safe and efficient charging, thereby enabling step-down of the PV voltage.

The control circuit incorporates the Perturb and Observe (P&O) MPPT algorithm, which operates periodically by perturbing the PV voltage and observing the resulting change in power. The PV voltage and current signals are passed through averaging blocks to remove high-frequency switching ripple and provide stable inputs for the MPPT, then generates a reference voltage V_{mppt} , which is compared with the measured PV voltage to produce an error signal. This error signal is processed by a proportional-integral (PI) controller to ensure that the PV voltage tracks the reference. The controller output is limited using a saturation block

with a limit of 1 and 0; the saturation block is compared with a repeating sequence (carrier signal) to produce a pulse width modulation (PWM) signal that drives the IGBT switch, allowing the buck converter to regulate the PV voltage and continuously track the MPPT.

The power stage includes an IGBT switch, an FWD, an inductor, and capacitors, which together regulate the energy transfer from the PV array to the battery. The PV array provides a variable DC voltage, which is stepped down to a suitable level for battery charging.

An input capacitor is used to reduce voltage ripple at the PV side. The IGBT acts as a main switch, controlling energy transfer through PWM operation of 20kHz; ideal IGBT switching was chosen in this simulation, which means switching losses are absent. The target of the current task is to demonstrate battery charging through PV array, not IGBT losses. An FWD provides a path for current when the switch is off, ensuring a continuous inductor current and protecting against spikes due to the inductor's stored energy. The inductor stores and releases energy to smooth the current, while the output capacitor reduces voltage ripple at the load. The battery acts as the load receiving regulated power from the buck converter. Voltage and current sensors are included to measure system variables for monitoring and control. Irradiance blocks of different values are used to simulate the power source for the PV array. Solver is used to control the step size.

2) Design a suitable inductance of the buck converter output inductor so that maximum peak-peak value of inductor ripple current is less than 2A. Comment on the effect of PWM frequency and the output inductance on the ripple current.

For a buck converter, the inductor ripple current is:

$$\Delta I_L = \frac{(V_{in} - V_{out})D}{Lf}; \text{ the buck converter is ideal, so } D = \frac{V_{out}}{V_{in}} = \frac{48}{150} = 0.32; \Delta I_L \leq 2 \text{ A.}$$

$$L \geq \frac{V_{out}(1-D)}{\Delta I_L f_s}; \quad L \geq \frac{48(1-0.32)}{2 \times 20000} \geq 816 \times 10^{-6} \text{ H}; L \geq 816 \mu\text{H}$$

The inductor value chosen for the simulation is L = 820 μH to keep the inductor current ripple below 2 A. This value is practical and matches the design requirement for real-world testing and experimentation.

The ripple current is inversely proportional to both switching frequency and inductance $\Delta I_L \propto \frac{1}{Lf_s}$, so increasing the inductance reduces the ripple current and vice versa. An increase in PWM frequency also reduces ripple current because the inductor is charged and discharged for a shorter time in each switching cycle, but it may increase switching losses in practice. A higher PWM frequency can reduce the required inductor size, resulting in a smaller circuit, lower cost, and faster response. The PWM frequency must be chosen within practical limits, considering both realistic operating constraints and component availability.

3) Specify suitable capacitance for C₁ in Figure 1.

A voltage ripple ΔV of 1% is assumed for both input (C_{in}) and output (C) capacitors to calculate their values. Although 1% ripple is conservative design assumption, way lower than typical practical values, and is selected to ensure enhanced filtering and stable performance.

$$\Delta V_{in} = 0.01 \times 150 = 1.5 \text{ V}; I_o - \text{output current}; I_o = \frac{P}{V_o} = \frac{150.6 \times 14.6}{48} = 45.81 \text{ A}; C_{in} \text{ or } C_1 = \frac{I_o D(1-D)}{\Delta V_{in} f_s} = \frac{45.8 \times 0.32 \times 0.68}{1.5 \times 20000} = \frac{9.97}{30000} = 3.32 \times 10^{-4} \text{ F} = 332 \mu\text{F};$$

The calculated input capacitance C₁ or C_{in} is equal to 330μF, representing the minimum value that is required to satisfy the selected ripple constraint. However, with a capacitance closer to 330 μF, the system becomes more robust to sudden changes in irradiance, which in our system would often occur due to the appearance of clouds or other weather changes.

$$\tau\text{-time constant}; \tau = R_{source} \times C_{in} = \frac{150.6}{14.6} \times 332 \mu = 3.41 \text{ ms};$$

$$\text{but if } C_{in} = 100 \mu\text{F, then; } \tau = \frac{150.6}{14.6} \times 100 \mu = 1.03 \text{ ms};$$

delay per step (3 × τ) = 10.23 ms and 3.09 ms, assuming during heavy cloud day 30% instantaneous change of 150.6 V = 45.2 V

, assuming $\Delta V_{\text{step}}=2$ - from MPPT design (Fig 13) $45.2/2=23$ steps that takes to retrack after 30% change in voltage: $T_{\text{retrack}} (300 \mu\text{F}) = 23 \times 10.23 = 235.29 \text{ ms}$,

$T_{\text{retrack}} (100 \mu\text{F}) = 23 \times 3.09 = 71.07 \text{ ms}$, which is approximately 3 times lower delay than $T_{\text{retrack}} (300 \mu\text{F})$. $C_{\text{in}}=100 \mu\text{F}$ is an optimal choice for filtering and quick P&O response. $C_{\text{in}}=100 \mu\text{F}$ also fits within design choice for P&O sampling time of 5ms which is mentioned in Table 2.2.

$\Delta V_o = 0.01 \times 48 = 0.48 \text{ V}$; $C = \frac{V_{\text{in}} D (1-D)}{8 L \Delta V_o f_s^2} = \frac{150 \times 0.32 \times (1-0.32)}{8 \times 820 \times 10^{-6} \times 0.48 \times (20000)^2} \approx 26 \mu\text{F}$. Assuming a 1% output ripple voltage, the required capacitance is approximately 26 μF . A higher practical value of 33 μF is selected to improve filtering and reduce ripple. 33 μF is an upper-band design choice and is realistic for implementation and experimental testing without introducing unnecessary system delay. $C=33 \mu\text{F}$.

4) Design and present the PV voltage controller, specify the controller parameters. Comment on the effect of the controller parameters on the control performance.

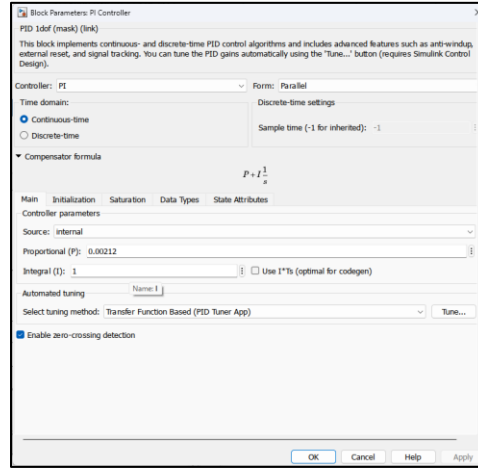


Figure 11. PI-proportional integral PV voltage controller

Fig. 11 illustrates a PI voltage controller used to regulate the PV array terminal voltage to track the reference voltage V_{mppt} generated by the P&O MPPT algorithm. The controller computes the error between the measured PV voltage and V_{pv} and the V_{mppt} and produces a signal to drive the IGBT switch connected to the buck converter.

The PI transfer function formula:

$$G(s) = K_p \frac{K_i}{s};$$

Using lecture-derived formulas:

$$\omega_0 = \frac{1}{\sqrt{L \times C_{\text{in}}}} = \frac{1}{\sqrt{820 \times 10^{-6} \times 100 \times 10^{-6}}} = 3496 \frac{\text{rad}}{\text{sec}}; \omega_c = \frac{\omega_0}{5} = \frac{3496}{5} = 699.2 \frac{\text{rad}}{\text{sec}}$$

$$K_p = \frac{D}{V_{\text{in}}} = \frac{0.32}{150.6} = 0.00212; K_i \leq \frac{K_p \times \omega_c}{5} = \frac{0.00212 \times 699.2}{5}; K_i \leq 0.296$$

The proportional gain K_p controls the speed of voltage error correction. If it is too large, it may cause the loss of regulation and oversaturation of the buck converter, while being too small may cause a delayed controller response to a voltage change. V_{pv} would lag behind V_{mppt} and not operate in the MPP range. The integral gain K_i eliminates steady-state error between V_{pv} and V_{mppt} , and if K_i is too large, it introduces overshoot and oscillation in the duty cycle, while too small allows steady-state error (sse) to persist, preventing the system from operating at MPP. Since V_{mppt} is updated every 5 ms by the P&O algorithm, both gains must be tuned such that V_{pv} settles to V_{mppt} within this time window, otherwise the P&O algorithm makes decisions based on unsettled measurements, leading to a miss-track or loss of power extraction efficiency.

The design formula gives $K_i \leq 0.296$ for $C_{in} = 100 \mu F$. However, $K_i = 1$ was selected in the simulation to achieve a faster voltage-tracking response. High K_i reduces the time it takes for V_{mppt} to converge to V_{pv} after each perturbation step. While this exceeds the calculated boundary, the 3-second simulation duration is too short for instability to develop. However, in real hardware, K_i should be limited to or set to 0.296 to ensure stable operation. $K_p = 0.00212$ was chosen for the simulation since the value is supported by the lecture-derived formula.

5) Design and present the PWM generator circuit for the buck dc-dc converter.

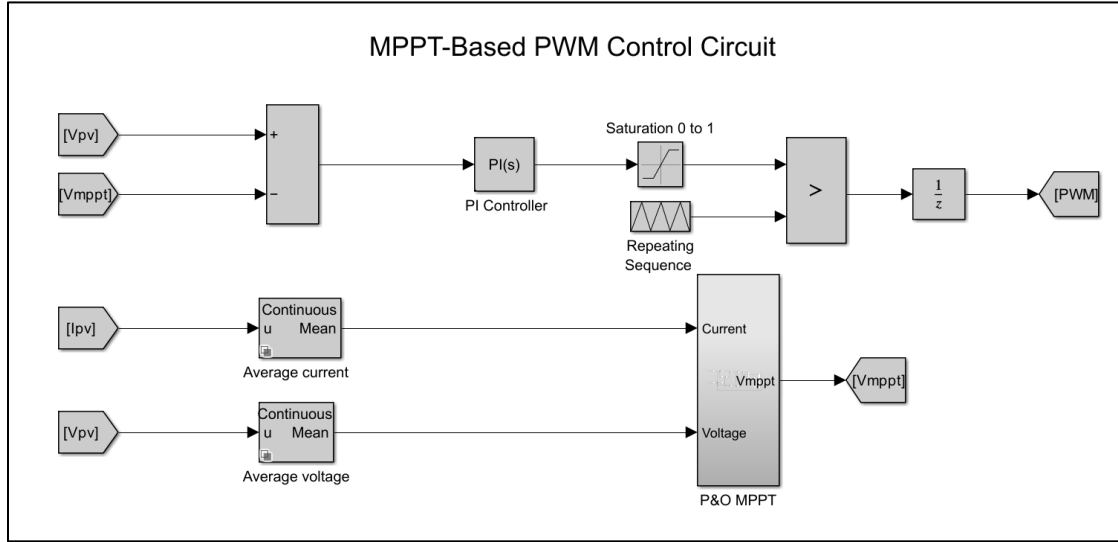


Figure 12. PWM-generator circuit and P&O MPPT block

Fig.12 illustrates the PWM-generator circuit and the P&O MPPT block. The PWM-generator produces a gate signal for the buck converter IGBT switch. It consists of a saturation block, a 20 kHz repeating sawtooth sequence, a comparator and a unit delay block. The PI controller output is first saturated between 0 and 1 to ensure a valid duty-cycle range, then compared with the sawtooth carrier waveform.

$$\text{Switching period: } T_{sw} = \frac{1}{f_{sw}} = \frac{1}{20000} = 50 \mu s; \text{ Duty cycle: } D = \frac{V_o}{V_{in}} = \frac{48}{150.6} = 0.32;$$

$$\text{IGBT is on for: } t_{on} = D \times T_{sw} = 0.32 \times 50 = 16 \mu s; \text{ IGBT is off for: } t_{off} = 50 - 16 = 34 \mu s$$

When the PI output exceeds the sawtooth the IGBT is on, otherwise the IGBT is off, producing a PWM signal of 20 kHz. The duty cycle equals to 0.32, giving an on time of 16 μs and off time of 34 μs per switching cycle. The unit delay block delays the signal until the next solution arrives.

6) Design and present the maximum power point tracking (MPPT) algorithm for the PV system so that the PV array can always output its maximum power to charge the battery under the given solar irradiances and temperatures. Specify the relevant parameters of the MPPT algorithm. Comment on the effect of the controller parameters on the system performance as well as the MPPT tracking efficiency.

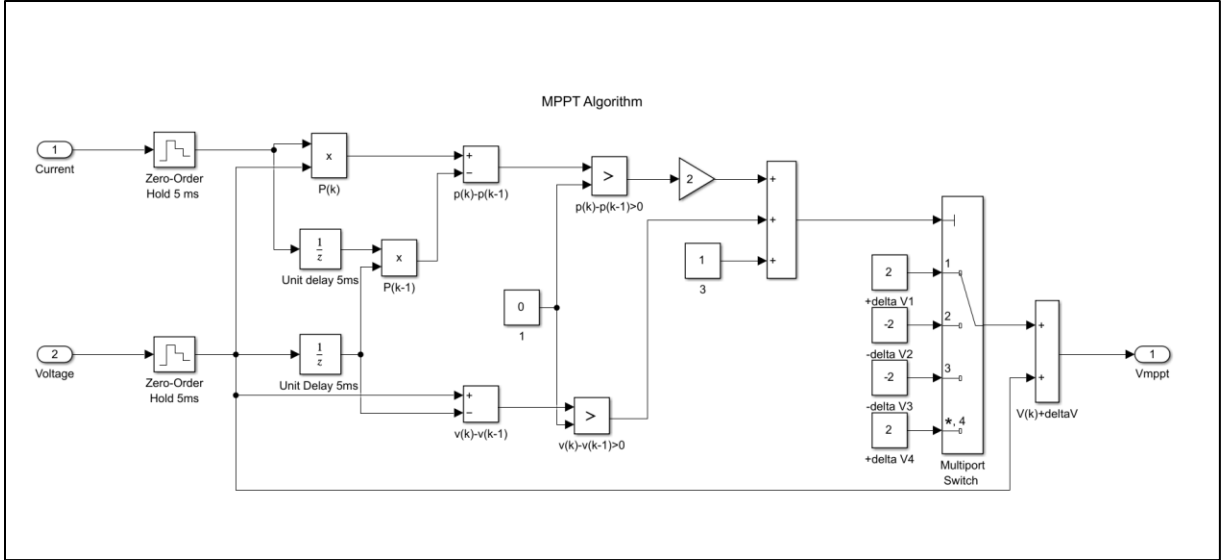


Figure 13. MPPT algorithm

Fig. 13 illustrates a P&O MPPT, designed for a PV system to bring the PV array closer to its MPP for maximum energy transfer and minimise losses. The controller measures PV voltage and current to calculate power, then compares it with the previous value. The P&O method observes the change in power: if the power increases, the perturbation continues in the same direction; if the power decreases, the direction is reversed, constantly tracking MPP. At each sampling instant the algorithm samples the PV voltage and current using zero-order hold blocks with a sample time of 5 ms. The instantaneous power is calculated as $P(k) = I(k) \times V(k)$ and compared against the previous power sample $P(k-1)$, this value then stored in 5 ms unit delay block. Similarly, the current voltage $V(k)$ is compared against the previous voltage $V(k-1)$. Based on the signs of $P(k) - P(k-1)$ and $V(k) - V(k-1)$, the multiport switch selects appropriate voltage perturbation from +2 V or -2 V which is added to the V_{pv} in order to achieve V_{mpp} .

Parameter	Value	Block
Sample time	5 ms	Zero-Order Hold
Unit delay	5 ms	Unit delay blocks
ΔV_{step}	± 2 V	Multiport Switch Inputs
Power comparison	$P(k) - P(k-1)$	Subtractor
Voltage comparison	$V(k) - V(k-1)$	Subtractor
Output	$V(k) + \Delta V$	Summing block

Table 2.2 Parameter table of P&O MPPT

The sample time of 5 ms is chosen to satisfy the capacitor settling requirement of $3\tau = 3.09$ ms for $C_{in} = 100$ μ F. Additionally, the 5ms sample time is well within the 71 ms retracking budget calculated for a 30 % irradiance step change, providing sufficient headroom for the algorithm to respond. A shorter sample time would cause mistracking in power comparison, while longer would slow transient response and reduce power tracking efficiency of the system. Perturbation Step Size $\Delta V = \pm 2$ V was optimal because it is large enough to reduce tracking time during transients and not too large to cause an overshoot and is larger than the voltage ripple theoretical calculation above of 1.5 V, so the system would classify 1.5 V as a noise, ensuring the MPPT algorithm distinguishes power changes from voltage ripple, preventing false perturbations. The perturbation size of ± 2 V exceeds the input voltage ripple of 1.5 V and was calculated in question 3). A smaller step would improve accuracy and reduce oscillation around V_{mpp} , but would increase retracking time after each irradiance change. A larger would reduce tracking time but increase overshoot. Overall, properly tuned parameters improve MPPT tracking efficiency by keeping the operating point close to the MPP with minimal oscillations.

2.2 1) Present the output voltage, output current and output power waveforms of the PV Array under the various solar irradiances. Comment on variations of PV array output voltage, output current and output power under different solar irradiances. Comment on the accuracy of the PV array output voltage, current and power by comparing the ideal/theoretical values of the PV array for the given conditions.

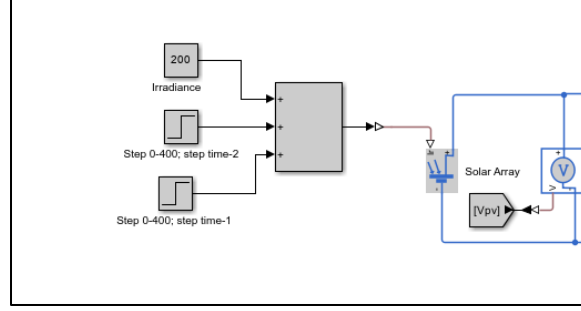


Figure 14 Modified MATLAB PV array for different solar irradiances (200 W×m⁻², 600 W×m⁻² and 1000 W×m⁻²)

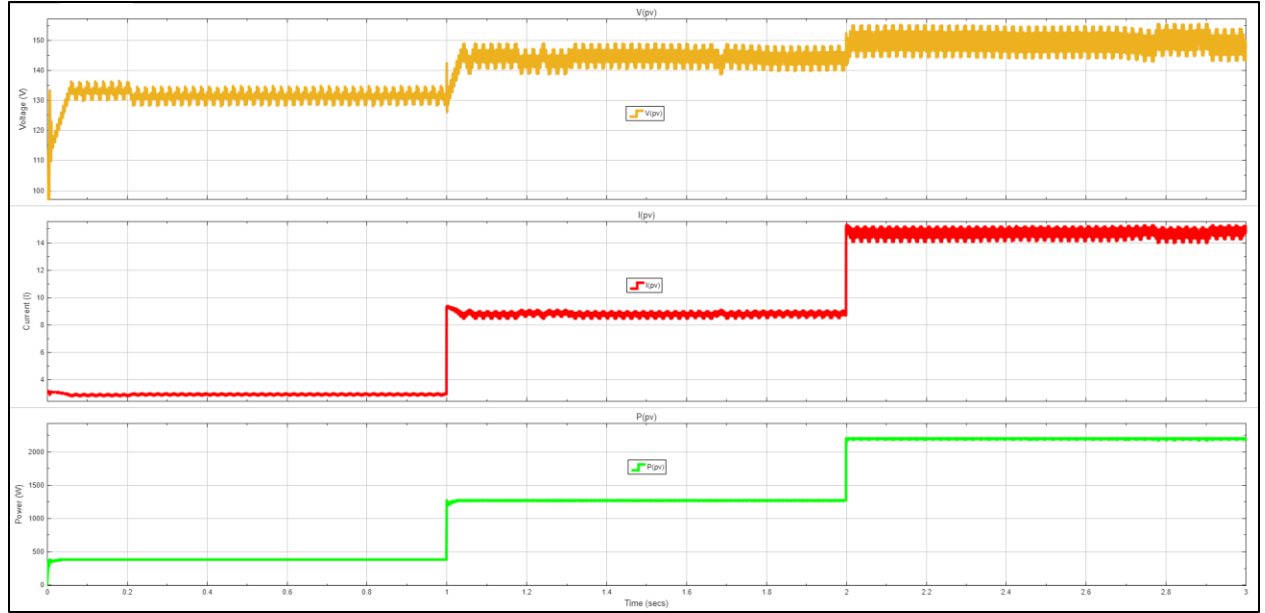


Figure 15 Output voltage, current and power waveforms of the PV array under the various solar irradiances ($C_{in} = 100 \mu F$)

Fig. 15 illustrates output power characteristics of the PV array under three different solar irradiances in a span of 3 seconds, 0 to 1(secs)- 200 (W×m⁻²), 1 to 2 (secs)- 600 (W×m⁻²), 2 to 3 (secs)- 1000 (W×m⁻²), while the device temperature remains constant at 25 °C. The step increase in irradiance corresponds to step rises in PV current (I_{pv}), power (P_{pv}), and in voltage (V_{pv}), demonstrating the direct dependence of PV output on solar irradiance. Using the zooming in MATLAB feature to estimate V_{mpp} at 200, 600 and 1000 (W×m⁻²) irradiances. $V_{mpp(200)} \approx 133V$; $V_{mpp(600)} \approx 145 V$; $V_{mpp(1000)} \approx 150 V$. Due to the absence of certain parameters such as I_0 and R_{sh} , the V_{mpp} is obtained from the steady state simulation results, while the current is calculated using the irradiance-dependent photocurrent relation. The aim of this simulation is to observe the effectiveness of the MPPT P&O system, not PV losses, so the simplified formula is used. The reference irradiance $G_{ref} = 1000 \text{ W} \times \text{m}^{-2}$ is used because it corresponds to standard test conditions (STC), at which the PV parameters, such as short circuit current, are defined, allowing the current to be scaled proportionally with irradiance. Analytical estimation of theoretical (th) values:

$$I_{pv(th)(200)} = 2 \times I_{sc} \frac{G}{G_{ref}} = 2 \times 7.95 \times \frac{200}{1000} = 3.18 \text{ A}; P_{(th)200} = V_{mpp(200)} \times I_{pv(th)(200)} = 133 \times 3.18 = 422.94 \text{ W}$$

$$I_{pv(th)(600)} = 2 \times I_{sc} \frac{G}{G_{ref}} = 2 \times 7.95 \times \frac{600}{1000} = 9.54 \text{ A}; P_{(th)600} = V_{mpp(600)} \times I_{pv(th)(600)} = 144 \times 9.54 = 1373.76 \text{ W}$$

$$I_{pv(th)(1000)} = 2 \times I_{sc} \frac{G}{G_{ref}} = 2 \times 7.95 \times \frac{1000}{1000} = 15.9 \text{ A}; P_{(th)1000} = V_{mpp(1000)} \times I_{pv(th)(1000)} = 150 \times 15.9 = 2385.0 \text{ W}$$

The maximum power at each irradiance is calculated using the corresponding V_{mpp} obtained from a steady-state simulation for that specific irradiance value. Simulated values obtained using the scope in MATLAB:

$$I_{pv(sim)(200)} = 2.94 \text{ A}; P_{pv(sim)(200)} = 385.86 \text{ W}; V_{mpp(sim)(200)} = 131.25 \text{ V}$$

$$I_{pv(sim)(600)} = 8.82 \text{ A}; P_{pv(sim)(600)} = 1270.08 \text{ W}; V_{mpp(sim)(200)} = 144 \text{ V}$$

$$I_{pv(sim)(1000)} = 14.75 \text{ A}; P_{pv(sim)(1000)} = 2197.75 \text{ W}; V_{mpp(sim)(1000)} = 149 \text{ V}$$

An alternative method for obtaining theoretical values is to use the MATLAB solar cell block digital datasheet, which provides an ideal maximum power P_{pv} of $200\text{-}1000 \text{ W}\times\text{m}^{-2}$. $P_{pv}(\text{datasheet})$ shows the ideal maximum power that can be generated by the cell, excluding parasitic resistances and circuit losses. These datasheet values offer a useful benchmark for validating simulated P_{pv} results against simulated (sim) values at each irradiance level:

$$P_{pv(datasheet)(200)} = 386.3 \text{ W}; \frac{P_{pv(sim)(200)}}{P_{pv(datasheet)(200)}} = \frac{385.86}{386.3} = 99.8 \%$$

$$P_{pv(datasheet)(600)} = 1270.1 \text{ W}; \frac{P_{pv(sim)(600)}}{P_{pv(datasheet)(600)}} = \frac{1270}{1270.1} = 99.9 \%$$

$$P_{pv(datasheet)(1000)} = 2205.9 \text{ W}; \frac{P_{pv(sim)(1000)}}{P_{pv(datasheet)(1000)}} = \frac{2197.75}{2205.9} = 99.6 \%$$

The simulated values are slightly lower than the theoretical values, with noticeable deviations in current and power, particularly at higher irradiance levels, attributable to MPPT ripple effects and non-ideal PV characteristics. Since the IGBT is modelled as an ideal switching device, the switching losses are negligible, and the remaining differences are primarily due to MPPT ripple and passive component losses, such as array resistance, diode forward resistance, inductor ESR, and capacitor parasitic resistance. The output voltage remains closely matched to the expected V_{mpp} , indicating that the MPPT controller is effectively maintaining operation near the MPP area despite these losses. It should be noted that $C_{in} = 330 \mu\text{F}$ would be more optimal choice for minimizing voltage, current and power ripple, as demonstrated in Fig.16. However, $C_{in} = 100 \mu\text{F}$ was selected to satisfy P&O sampling constraint and ensure an efficient MPPT P&O tracking algorithm, rather than to achieve the smoothest possible waveform. For experimental implementation, $330 \mu\text{F}$ would be a better alternative for improved power transfer continuity and reduced overall ripple.

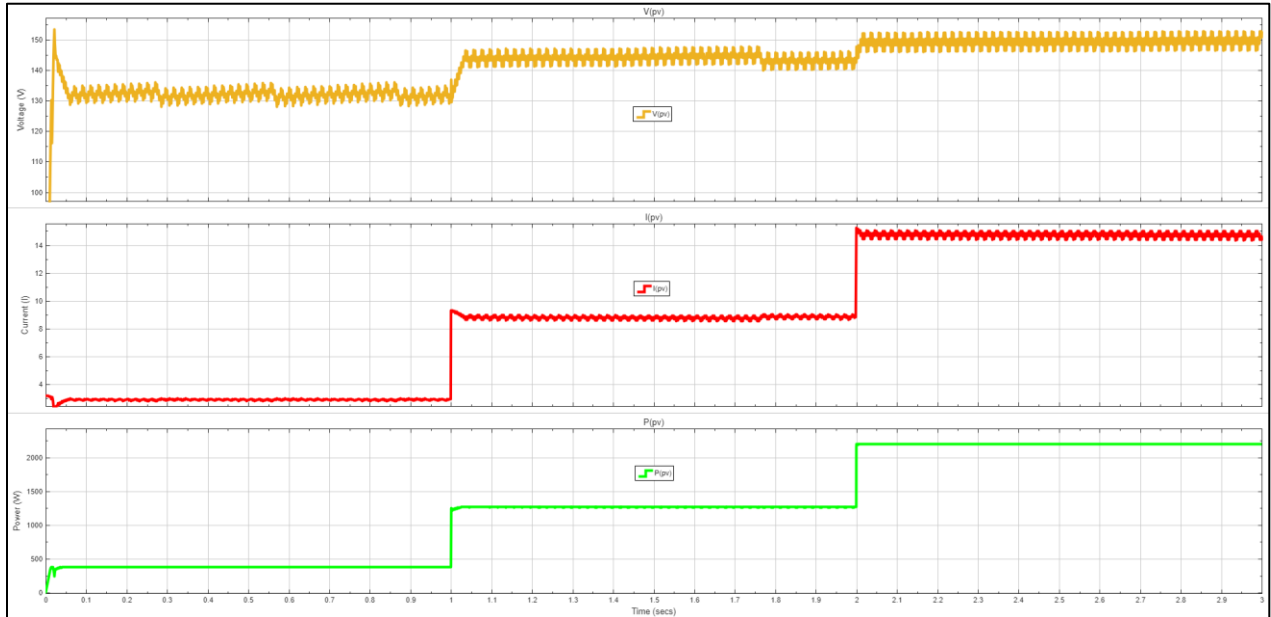


Figure 16 Output voltage, output current and output power waveforms under ($C_{in} = 330 \mu\text{F}$)

2) Comment on the performance of MPPT controller and effect of the MPPT parameter on the PV output voltage, current and power (transient response and steady state performance). Estimate the MPPT efficiency by comparing the theoretical values.

The MPPT controller maintains operation near the MPP. The PV voltage shows the largest ripple, the current has smaller ripple, and the power has the least ripple. This is because P&O directly perturbs the PV voltage towards MPP; the current response is naturally smoother due to the inductance in the buck converter, which prevents rapid current changes. Since the output power is the product of voltage and current, small variations in each partially cancel, making power relatively insensitive to individual component ripple. The MPPT demonstrates good performance, effectively tracking MPP with a fast transient response to irradiance changes and stable steady-state operation near V_{mpp} . The perturbation step size governs the speed-accuracy trade-off: a larger step gives faster tracking after irradiance changes but increases steady-state error (SSE), whereas a smaller step improves steady-state accuracy but risks significant energy loss during irradiance transitions, which are frequently encountered in solar systems.

MPPT efficiency is calculated using-[2]: $\eta_{MPPT} = \frac{P_{sim}}{P_{theoretical} \text{ or } P_{datasheet}} \times 100 \%$, where P_{sim} is the simulated power value, $P_{theoretical}$ is the theoretical upper-bound value derived from analytical calculations, and $P_{datasheet}$ is the ideal maximum power obtained from the MATLAB Simscape solar cell block datasheet.

Results:

$$200 \text{ W} \times \text{m}^{-2}: \eta_{theoretical} = \frac{385.86}{422.94} \times 100 = 91.23 \% ; \eta_{datasheet} = \frac{385.86}{386.3} \times 100 = 99.8 \%$$

$$600 \text{ W} \times \text{m}^{-2}: \eta_{theoretical} = \frac{1270}{1373.76} \times 100 = 92.4 \% ; \eta_{datasheet} = \frac{1270}{1270.1} \times 100 = 99.9 \%$$

$$1000 \text{ W} \times \text{m}^{-2}: \eta_{theoretical} = \frac{2220}{2385} \times 100 = 93 \% ; \eta_{datasheet} = \frac{2197.75}{2205.9} \times 100 = 99.6 \%$$

The theoretically based efficiencies are 91.2 %, 92.4 %, and 93 % at $200 \text{ W} \times \text{m}^{-2}$, $600 \text{ W} \times \text{m}^{-2}$, and $1000 \text{ W} \times \text{m}^{-2}$, respectively, giving an overall average of 92.2 %. The datasheet-based efficiencies were 99.8 %, 99.9 %, and 99.6 %, giving an overall average of 99.8 %. The inability to reach 100% datasheet efficiency is attributed to conduction losses and parasitic circuit effects, since IGBT switching losses are disabled in the MATLAB simulation. The gap between theoretical values is an upper-bound estimate under ideal conditions, whereas datasheet values represent the true maximum power limited by the physical characteristics of the PV cell.

The simulated PV power closely matches the total available power P_{max} , achieving nearly 100 % of the datasheet value, confirming the MPPT controller's efficiency. Therefore, the overall system, including the buck converter and control components, demonstrates high performance and successful extraction of maximum power.

3) Estimate the duty cycle in steady-state operation of the buck converter, present the duty cycle as well as PWM waveforms during the simulation. Comment its variation under different solar irradiances.

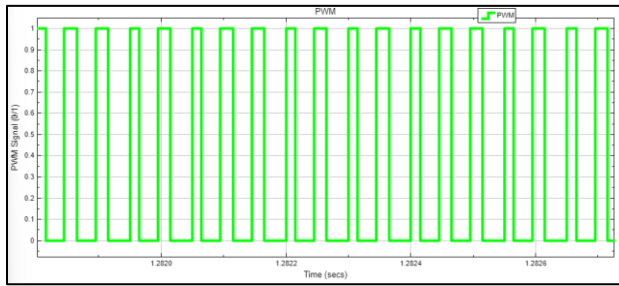


Figure 17 PWM waveform (zoomed in)

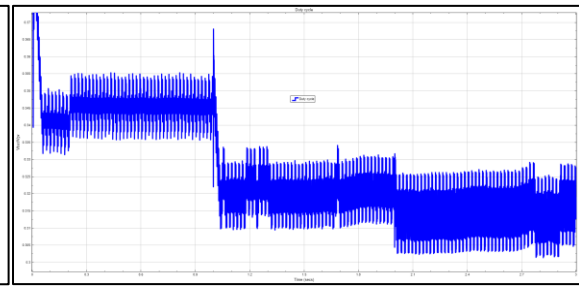


Figure 18 Duty cycle (full view)

Fig. 17 and Fig. 18 illustrate the PWM waveform at higher magnification and the corresponding duty cycle, respectively. The PWM waveform was obtained using a division block assuming an ideal buck converter. $D = \frac{V_{battery}}{V_{pv}} = \frac{V_{out}}{V_{in}}$. Theoretical duty cycle:

$$\text{At } 200 \text{ W} \times \text{m}^{-2}: D = \frac{48}{133} = 0.361 ; \text{At } 600 \text{ W} \times \text{m}^{-2}: D = \frac{48}{144} = 0.331 ; \text{At } 1000 \text{ W} \times \text{m}^{-2}: D = \frac{48}{150} = 0.320$$

The theoretical duty cycle was estimated using the ideal buck converter relationship; the simulated duty cycle is slightly lower and shows oscillations around these theoretical values due to MPPT perturbation and non-ideal circuit behavior, although both follow the same decreasing trend with increasing irradiance. Since the switching frequency is constant, the duty cycle is proportional to the IGBT conduction time. As irradiance increases, V_{pv} rises, requiring a lower duty cycle to maintain the target output voltage: Irradiance $\uparrow \rightarrow V_{pv} \uparrow \rightarrow D \downarrow \rightarrow$ narrower pulses $\rightarrow$ shorter IGBT conduction time.

Fig.18 shows a noticeable decrease in duty cycle with each irradiance step increase. The decrease in duty cycle from 600 to 1000 $W \times m^{-2}$ is more moderate than from 200 to 600 $W \times m^{-2}$, which is consistent with the PV array - V_{mpp} increases less steeply at higher irradiance levels, resulting in a smaller change in a duty cycle.

4) Present the battery input voltage and current as well as state-of-the-charge waveforms. Comment on the value and quality of battery charging voltage and current.

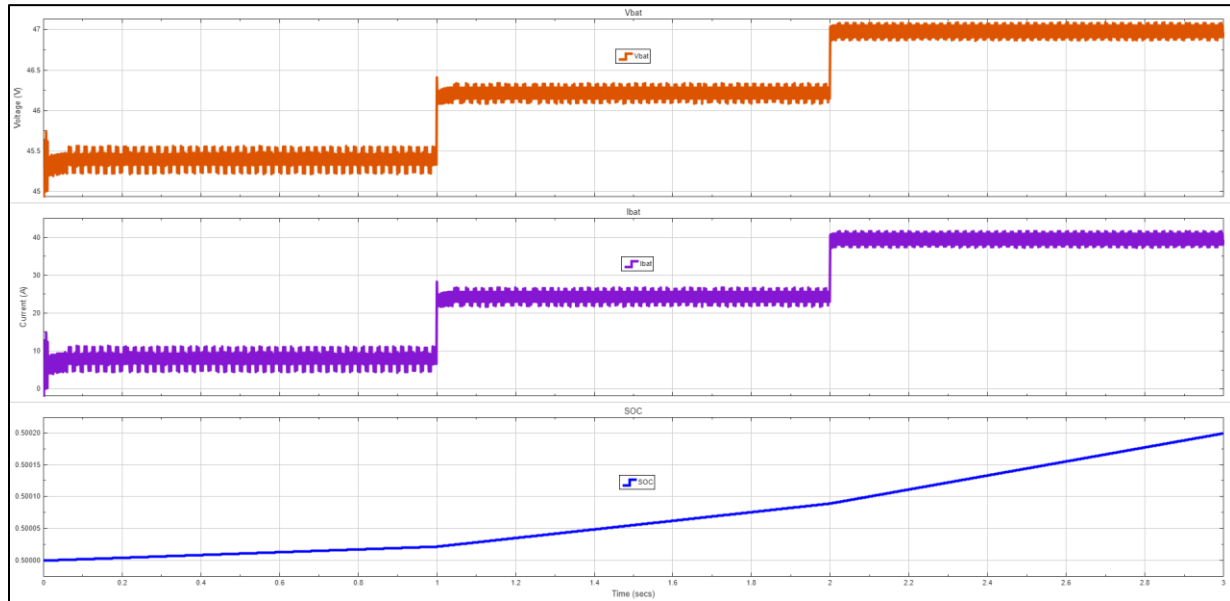


Figure 19 Battery input voltage, current and state of the charge (SOC) waveforms

Fig.19 illustrates that SOC increases slightly from 0.5 to 0.5002 over the simulation period, indicating that the battery charging is occurring, albeit slowly, due to the short 3-second simulation window and large battery capacity, the rate of state of charge remains small $\Delta SOC = 0.02 \%$. The battery voltage V_{bat} remains relatively stable at 44-46 V, increasing corresponding with irradiance step increase at $t = 1$ sec and $t = 2$ sec. The voltage ripple is small at $\sim 0.5 - 1$ V, indicating tight regulation, whereas current ripple is slightly bigger with ± 5 A, due to switching effects of the DC-DC step down converter (energy storage in the inductor, as well as charging and discharging of the capacitor). Current ripple is marginally larger, which is due to switching action of the DC-DC buck converter- specifically energy storage and release in the inductor and the charge or discharge of the capacitor during turn off and turn on cycles. SOC increases continuously throughout the simulation, with the rate becoming noticeably steeper following each irradiance step, reflecting the increased power delivered to the battery at higher irradiance levels. Finally, the battery is charged with voltage that is well-regulated and acceptably smooth current, confirming stable and efficient charging behavior that is consistent with the DC-DC buck converter.

2.2 & 2.3

Simulate the designed simulation circuit and its controller for the given cell temperatures: from 0 to 1 second - 25 ($^{\circ}C$), 1 to 2 second - 45 ($^{\circ}C$), from 2 to 3 second - 65 ($^{\circ}C$). The solar irradiance always remains at constant value of 1000 ($W \times m^{-2}$). Carry out the simulation for 3 seconds. Present the output voltage, output current and output power waveforms of the PV Array under the various temperature. Comment on variations of PV array output voltage, output current and output power under different solar cell temperatures. Comment on the accuracy of the PV array output voltage, current and power by comparing the ideal (theoretical) values of the PV array for the given conditions.

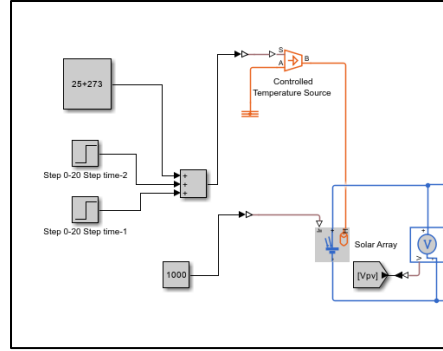


Figure 20 Modified PV array for different temperatures

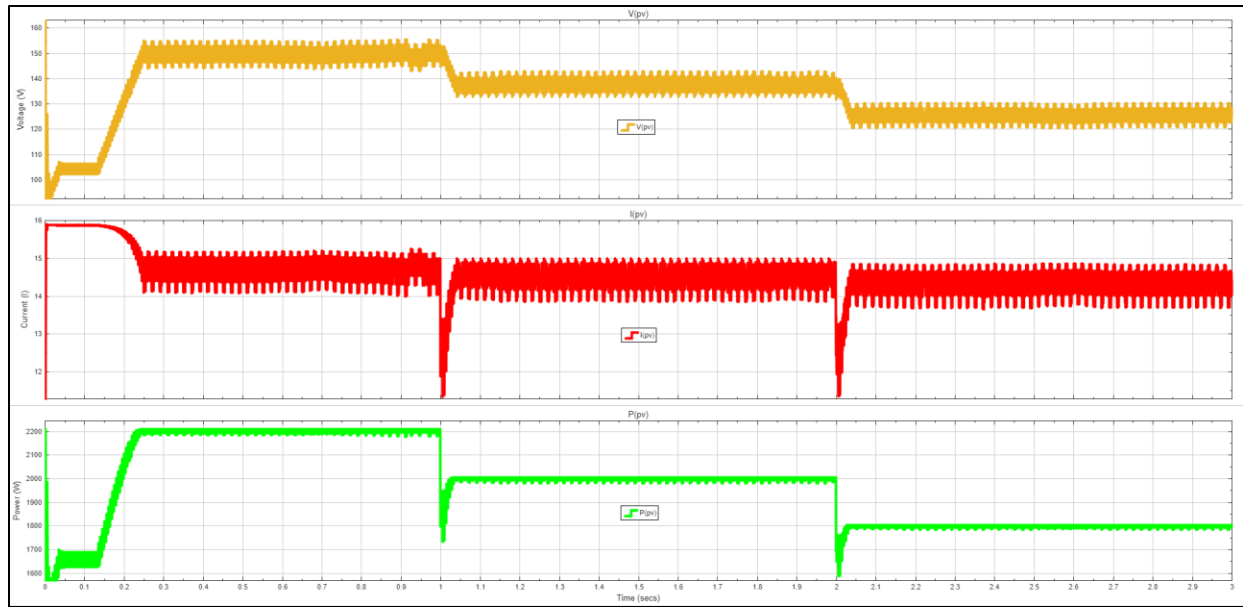


Figure 21 Output voltage, output current, and output power waveforms of the PV Array under the various temperatures (25 °C, 45°C, 65 °C)

The aim of this simulation is to illustrate the effects of temperature increase on PV voltage, current and power. Fig. 21 clearly indicates that the photovoltaic voltage V_{pv} decreases as temperature increases, while the PV current I_{pv} decreases only slightly with noticeable transient dips during each temperature step. The PV power undergoes a decrease as the temperature increases, since $\downarrow P_{pv} = \downarrow V_{pv} \times I_{pv}$ decreases accordingly. The power drops are around 250 W with each temperature step. The temperature increases in steps: from 0 to 1 seconds - 25 °C, 1 to 2 seconds - 45 °C, from 2 to 3 seconds - 65 °C, while the solar irradiance remains constant across all 600 cells at $1000 \text{ W} \times \text{m}^{-2}$, and the simulation runs for 3 seconds. The temperature source starts at 298 K (room temperature) and increases by 20 K which is equivalent to 20 °C at each step. Fig. 21 simulates a gradual temperature increase in the cell; however, it is assumed that all six hundred individual cells experience uniform temperature change, which simplifies the model, but may slightly overestimate performance compared to real systems with non-uniform temperature distribution. The general trend indicates a loss of approximately 200 W per 20 °C rise, equivalent to 10 W per °C in this system. MATLAB is fitted with a data sheet for the solar cell block for power readings under different temperatures, using as a reference for comparison with simulated results:

T(°C)	V_{pv} (Volt)	I_{pv} (Ampere)	P_{pv} (Watt)	V_{max} (Volt)	I_{max} (Ampere)	P_{max} (Watt)
25	150.0	14.7	2205.0	148.2	14.9	2205.9
45	137.8	14.5	1998.1	136.1	14.7	2002.7
65	125.5	14.3	1797.5	126.9	14.2	1801.7

Table 2.3 Theoretical datasheet values ($V_{\max}$, $P_{\max}$, and $I_{\max}$) and simulated values (V_{pv} , I_{pv} , and P_{pv}) under different temperatures during constant irradiance of $1000 \text{ W} \times \text{m}^{-2}$

The simulated results in Table 2.3 closely match datasheet theoretical values with deviations within approximately 2 %, confirming high model accuracy. Minor instances where I_{pv} and V_{pv} exceed datasheet maximum values are attributable to MPPT perturbation, as the operating point oscillates during operation rather than remaining fixed at MPP. This pattern does not significantly impact overall performance.

1) Comment on the performance of MPPT controller by comparing its actual output power with its theoretical values under different temperatures.

Using a formula to calculate efficiency at each temperature stage: $\eta_{\text{mppt}} = \frac{P_{pv}}{P_m} \times 100$, where P_{pv} -simulated values, and P_m - maximum datasheet values of the cell.

$$25^\circ\text{C}: \eta_{\text{mppt}} = \frac{2205.0}{2205.9} \times 100 = 99.96 \%$$

$$45^\circ\text{C}: \eta_{\text{mppt}} = \frac{1998.1}{2002.7} \times 100 = 99.77 \%$$

$$65^\circ\text{C}: \eta_{\text{mppt}} = \frac{1797.5}{1801.7} \times 100 = 99.43 \%$$

Overall performance is 99.72 %, signifying successful operation of the P&O MPPT controller and optimal power extraction under varying temperatures. As the temperature increases, both simulated and theoretical output power decrease, which is consistent with expected PV behavior. The small deviation from the $P_{\max}$ value is caused by MPPT perturbation around the MPP, PV array resistance, and DC-DC step-down buck converter operation losses. The result above indicates that the controller remains robust despite temperature variation with no significant degradation in tracking accuracy.

2) Estimate the duty cycle in steady-state operation of the buck converter and present the duty cycle waveform during the simulation. Comment its variation under different solar cell temperatures.

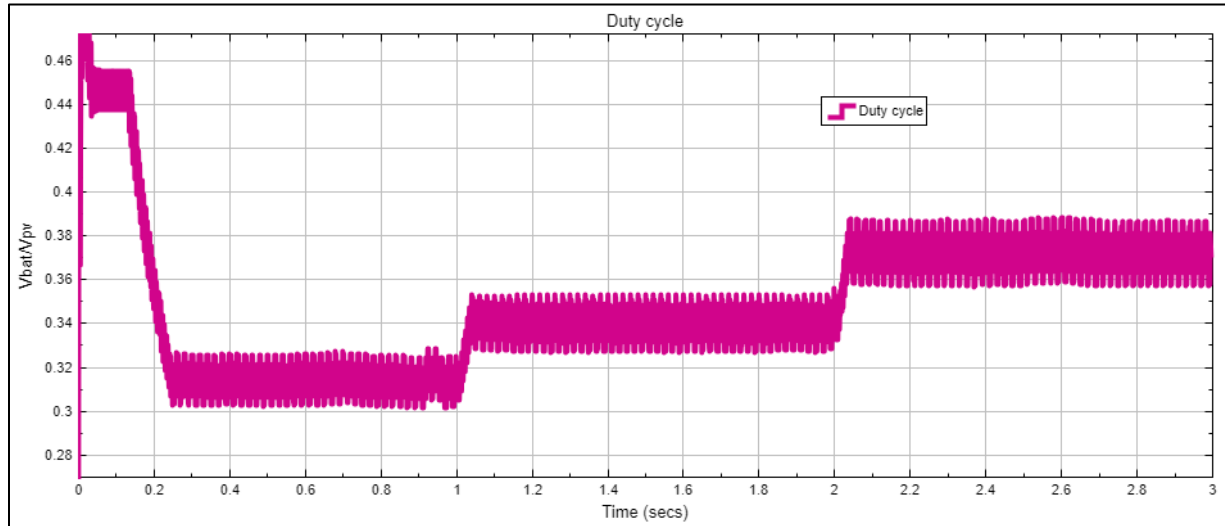


Figure 22 Duty cycle waveform under varying temperatures and at a constant irradiance of $1000 \text{ W} \times \text{m}^{-2}$

Estimating steady-state duty cycle estimation with zoom feature in MATLAB:

25 °C (0-1 s): $D \approx 0.315$

45 °C (1-2 s): $D \approx 0.340$

65 °C (2-3 s): $D \approx 0.375$

As temperature increases, bandgap energy decreases, narrowing reduces V_{oc} , and since $V_{mpp} = 0.8 \times V_{oc}$, V_{pv} decreases proportionally. $\uparrow D = \frac{V_{bat}}{V_{pv}}$, so reduction in V_{pv} causes an increase in D to maintain the output voltage: Temperature $\uparrow \rightarrow V_{pv} \downarrow \rightarrow D \uparrow$. The results indicate that each 20°C rise corresponds to approximately a 10% increase in duty cycle. Fig.22 also illustrates a short transient after each temperature step before settling to a new steady state value.

3) Present the battery input voltage and current as well as state-of-the-charge waveforms. Comment on the value and quality of battery charging voltage and current.

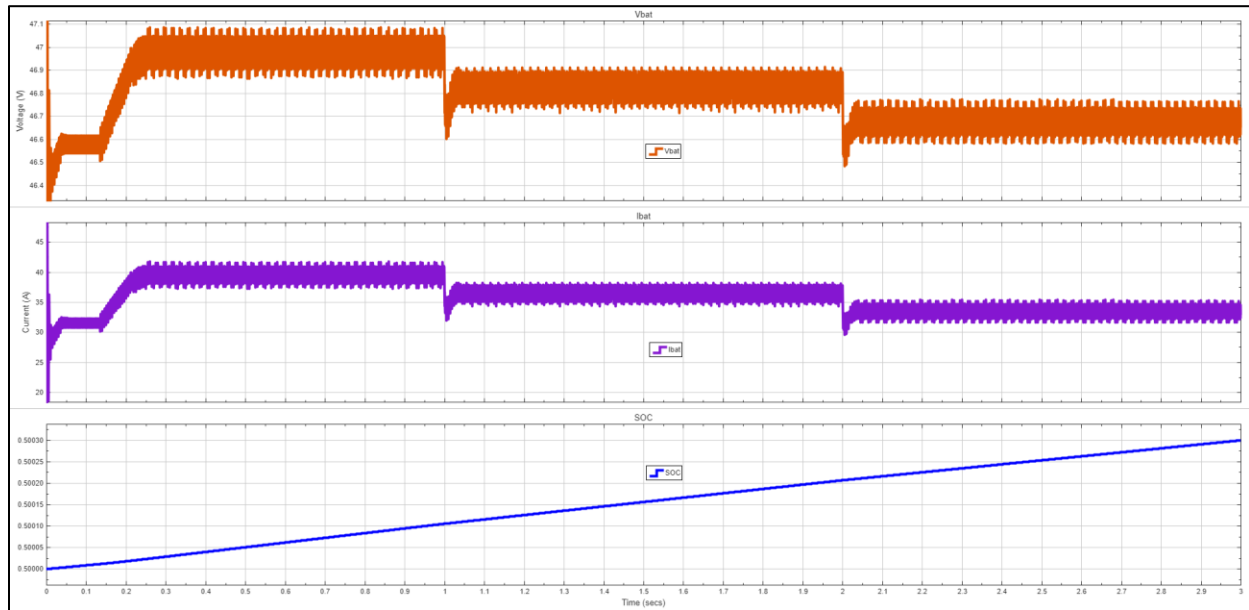


Figure 23. Waveforms of battery input voltage, current, and SOC

Firstly, the battery charging voltage remains approximately 46.6-47.1 V, close to the nominal battery voltage of 48 V, indicating good regulation by the buck converter. The voltage exhibits a small ripple with each temperature rise. The current is significantly higher under conditions of constant irradiance $1000 \text{ W}\times\text{m}^{-2}$, rather than varying irradiance of $200\text{-}600\text{-}1000 \text{ W}\times\text{m}^{-2}$, even at the increased temperature conditions, the lowest value of I_{bat} ($\sim 7.5 \text{ A}$) in Fig 19 is approximately four times smaller than in Fig. 23 - I_{bat} ($\sim 33.7 \text{ A}$). Such a difference in current is also reflected on the SOC graph, which ends at 0.50030 at Fig. 23, but at Fig. 19 the charge ends at 0.50020, which means the current setup under different temperature conditions is more beneficial due to higher SOC. SOC rate in Fig.23 is 0.06 % per second from total battery charge whereas previous one was 0.04 % per second in Fig.19. Overall, both input current and input voltage in battery are decreasing with increase in temperature of the PV array, however due to the constant irradiance of $1000 \text{ W}\times\text{m}^{-2}$ the battery charges faster, due to increased charging current, even when the temperature conditions are increasing, but in reality constant high temperature could possibly shorten lifespan of the battery, physical limitations are also important part of PV array when it comes to maintenance and cost efficiency. The quality of the battery charging is improved, and the amount of current and voltage used in charging also increases, but the ripple of voltage and battery current has more ripple under varying temperature conditions.

III IGBT power losses and thermal design

3.1) If the IGBT switch is implemented by using IKW30N60H3 IGBT device (Data sheet uploaded in canvas/Assignment Overview), estimate the IGBT power loss under the given condition (solar irradiance of $1000 \text{ W}\times\text{m}^{-2}$) and Device junction temperature is 65°C . Calculate the percentage of this power loss against the power delivered. Comment on the effect of PWM switching frequency and duty cycle on the power losses

IGBT power loss at 65°C ? $P_{IGBT(65^\circ\text{C})} = P_{cond} + P_{sw} = P_{diss}$

$D = 0.372$, $P_{\text{bat}} = 1575 \text{ W}$; $V_{\text{bat}} = 46.675 \text{ V}$; $I_{\text{bat}} = 33.744 \text{ A}$. $V_{\text{pv}} = 125.5 \text{ V}$ - These values were obtained using MATLAB scope at 65 °C and irradiance of $1000 \text{ W} \times \text{m}^{-2}$

$$f_{\text{sw}} = 20 \text{ kHz}$$

Obtained via IKW30N60H3 datasheet:

$$\text{At } 175 \text{ }^{\circ}\text{C}: V_{\text{ce}} = 2.50, E_{\text{ts}} = 1.72 \text{ mJ}$$

$$\text{At } 125 \text{ }^{\circ}\text{C}: V_{\text{ce}} = 2.30 \text{ V}$$

$$\text{At } 25 \text{ }^{\circ}\text{C}: E_{\text{ts}} = 1.38 \text{ mJ}$$

$$P_{\text{cond}} = V_{\text{CE(sat)}} I_{\text{c}} D$$

In a buck converter, the IGBT carries the output current during the on time, so:

$$I_{\text{c}} \approx I_{\text{bat}} = 33.744 \text{ A}$$

Using linear interpolation to find V_{ce} : $y = mx + c$, where $T = V_{\text{ce}}(\text{m}) + c$

$$m = \frac{125-175}{2.30-2.50} = 250$$

$$125 = 2.3 \times 250 + c \Rightarrow c = -450$$

$$65 = 250 \times V_{\text{ce}} - 450$$

$$V_{\text{ce}} = 2.06 \text{ V at } 65 \text{ }^{\circ}\text{C}$$

$$P_{\text{cond}} = V_{\text{CE(sat)}} I_{\text{c}} D = 2.06 \times 33.744 \times 0.372 = \underline{\underline{25.86 \text{ W}}}$$

$$P_{\text{sw}} = E_{\text{sw}} \times f_{\text{sw}}$$

Using linear interpolation to find E_{sw} at 65 °C: $y = mx + c$, where $T = E_{\text{sw}}(\text{m}) + c$

$$m = \frac{175-25}{1.72-1.38} = 441.18$$

$$25 = 441.18 \times 1.72 + c \Rightarrow c = -733.83$$

$$65 = E_{\text{sw}} \times 441.18 - 733.83$$

$E_{\text{sw}} = 1.811 \text{ mJ}$ - this value was obtained from the datasheet with conditions of $I_{\text{c}} = 30 \text{ A}$ and $V_{\text{ce}} = 400 \text{ V}$, so this value must be converted to the operating point of $I_{\text{c}} = 33.744 \text{ A}$ and $V_{\text{ce}} = 125.5 \text{ V}$:

$$E_{\text{sw}} = 1.811 \times \frac{33.744}{30} \times \frac{125.5}{400} = 0.639 \text{ mJ}$$

$$P_{\text{sw}} = f_{\text{s}} \times E_{\text{sw}} = 20000 \times 0.639 \times 10^{-3} = \underline{\underline{12.78 \text{ W}}}$$

$$P_{\text{diss}} = P_{\text{cond}} + P_{\text{sw}} = 12.78 + 25.86 = \underline{\underline{38.64 \text{ W}}}$$

$$P_{\text{IGBT loss}} = \underline{\underline{38.64 \text{ W}}} \text{ at } 65 \text{ }^{\circ}\text{C} \text{ and } 1000 \text{ W} \times \text{m}^{-2} \text{ irradiance}$$

$$P_{\text{delivered}} = P_{\text{bat}} = 1575 \text{ W}$$

$$\% \text{loss} = \frac{38.64}{1575} \times 100 = \underline{\underline{2.45 \%}}$$

$P_{\text{sw}} = E_{\text{sw}} \times f_{\text{sw}}$, so increasing PWM frequency increases switching loss almost linearly, the more times the device switches per second the higher its switching losses. However, increasing switching frequency also reduces the required inductance and capacitance for the same ripple target, representing a trade-off between switching losses and passive component size and system speed.

$\uparrow P_{\text{cond}} = V_{\text{CE(sat)}} I_c D \Rightarrow P_{\text{diss}} \uparrow$ Increasing the duty cycle increases conduction losses, since the device is conducting for a longer period of time, the IGBT remains ON for longer in each switching period. Dissipation losses increase with switching losses, so an increased duty cycle makes the IGBT energy inefficient and results in higher energy losses. On the opposite side, increasing the duty cycle increases the on-time, leaving less time off and non-conducting.

3.2) Assume the ambient temperature is 25 degree, based on the power loss estimated from 3.1, work out the device junction temperature without installing heatsink.

Without heatsink: $T_j = T_a + P_{\text{loss}} \times R_{\text{th}(j-a)}$

$$T_j = 25 + (38.64) \times 40 = 1570.6 \text{ } ^\circ\text{C}$$

With heatsink: $T_j = T_a + P_{\text{loss}} \times R_{\text{th}(j-a)}$, where $R_{\text{th}(j-a)}$ is junction to ambient

$$T_j = T_a + P_{\text{loss}} (R_{\theta(j-c)} + R_{\theta(c-s)} + R_{\theta(s-a)})$$

$R_{\text{th}(j-a)}$ = junction-to-ambient (without heatsink)

$R_{\theta(j-c)}$ = junction-to-case thermal resistance

$R_{\theta(c-s)}$ = case-to-heatsink thermal resistance

$R_{\theta(s-a)}$ = heatsink-to-ambient thermal resistance

The data sheet does not provide $R_{\theta(c-s)}$ and $R_{\theta(s-a)}$, so the calculation below is non-ideal.

$$R_{\theta(j-c)} = 0.4 \text{ } ^\circ\text{C/W}, T_j = 25 + (38.64) (0.4) = 40.6 \text{ } ^\circ\text{C}$$

For operation without a heatsink and no controlled case cooling, there is no defined thermal path through the case; therefore, heat flows directly from the junction to ambient. As a result, the appropriate parameter is junction-to-ambient thermal resistance, which exceeds the physical limit of the device by approximately 1400 $^\circ\text{C}$, meaning the device cannot operate without a heatsink. If there were a heatsink installed, using $R_{\theta(j-c)} = 0.4 \text{ K/W}$ gives $T_j = 40.6 \text{ } ^\circ\text{C}$, which is below the device's threshold temperature of 175 $^\circ\text{C}$; however, this assumes a controlled case temperature and is not valid if the system does not have any heatsink installed. This device is inoperable without a controlled heatsink and a value of 1570 $^\circ\text{C}$ is non-physical.

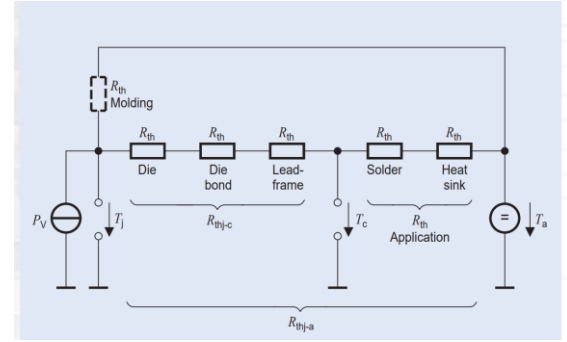


Figure 24. Thermal equivalent circuit. Figure taken from- [4]

IV Conclusion

Overall, the PV battery charging system was successfully designed and validated through simulation. The PV array configuration achieved the required output levels, and the I-V and P-V characteristics illustrated expected behavior under varying irradiance and temperature conditions. The result of the second simulation confirmed that increasing irradiance increases maximum power, while increasing temperature reduces the PV voltage and leads to a noticeable decrease in output power. MPPT-controlled buck converter maintained operation close to the maximum power point under varying tested conditions, ensuring efficient power and charge transfer from the PV array to the battery. The achieved simulation results were compared to theoretical calculations and MATLAB data-sheet specifications; the effectiveness of MPPT ranges from 91 %- theoretical to 99.9 % - PV array datasheet. The PWM and duty cycle waveforms illustrated inverse proportionality to PV voltage under increased irradiance and different temperature conditions.

The battery charging was successful and demonstrated an increase in the rate of charge (0.04 %) with an increase in irradiance; however, an increase in temperature caused a higher rate of charge (0.06 %) in case 3 seconds of simulation time. SOC illustrated a continued increase for all simulated conditions, demonstrating reliable charging performance.

The IGBT power loss and thermal analysis showed low conduction and switching power losses compared to the delivered power (2.45 %). Operation without a heatsink is impossible and would result in extremely high junction temperature that is incompatible with device operation. In conclusion, the simulation results confirm successful system design, effective MPPT tracking, and stable battery charging, and highlight the importance of thermal management in power electronic systems.

V References

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- [4] Infineon Technologies ltd., "Thermal Resistance Theory and Practice," Infineon Technologies AG, Munich, Germany, App. Note AN 2008-03, 2008.

Appendix A:

Solar array specification for task 2

Battery specification for task 2

Appendix B:

Component code	Parameter
Input Capacitor (C_{in}) - 1410454C	100 μ F
Output capacitor (C) - 1433393C	33 μ F
Inductor (L)* - 1410516C	1 mH
IGBT Switch - IKW30N60H3	600 V 30 A
Lithium battery	48V, 2-5Ah
STPS3L60Q_Diode	Switching diode
Solar Panel	$V_{mpp} \approx 30V$, $I_{mpp} \approx 7.5A$

Table 3.1 Specifications for experimental implementation

*- as 820 μ H is not available from current component list the nearest standard value of 1.0 mH is selected which exceeds the minimum requirement and provides additional margin